

AiNeura Retrieval Analysis

Context / Memory Retrieval Evaluation — 40 Queries · 160 Memories · Production Pipeline

● Run: 2026-09-02T04:58:03Z ● topK=8 · vector=0.5 · lexical=0.2 · importance=0.2 · recency=0.1

● Topical Penalty: DISABLED ● Tests passing: 797/797

1 Aggregate Metrics

QUERIES

40

EXPECTED

104

FETCHED

320

TRUE POSITIVES

54

FALSE POSITIVES

266

FALSE NEGATIVES

50

PRECISION

16.9%

RECALL

51.9%

F1 SCORE

25.5%

● Critical Finding

266 out of 320 total fetched memories are false positives — **83.1% of all retrieved memories are irrelevant.** The system correctly finds the relevant memories (recall 51.9%) but cannot filter out the noise (precision 16.9%). Five memories alone account for 45.1% of all false positives.

QUERY ID	QUERY	EXPECTED	FETCHED	TP	FP	FN	PRECISION	RECALL	F1	MAIN ISSUE
Q09	What tools and frameworks do I use for testing?	2	8	0	8	2	0.000	0.000	N/A	Conceptual confusion
Q21	What are my current engineering goals for this quarter?	3	8	0	8	3	0.000	0.000	N/A	Sess: failure
Q33	What is my health and fitness routine?	5	8	0	8	5	0.000	0.000	N/A	Sam: clus
Q36	What do I know about quantum computing?	1	8	0	8	1	0.000	0.000	N/A	Low targ
Q37	Tell me about my personal interests and hobbies outside work.	4	8	0	8	4	0.000	0.000	N/A	Hob: disp
Q38	What is the recipe for making sourdough bread?	1	8	0	8	1	0.000	0.000	N/A	Targ: impo neve
Q39	What are all my upcoming important	5	8	0	8	5	0.000	0.000	N/A	Cros: agg

QUERY ID	QUERY	EXPECTED	FETCHED	TP	FP	FN	PRECISION	RECALL	F1	MAIN CLUSTER
	deadlines before year-end?									
Q32	What are my diet and nutrition habits?	5	8	1	7	4	0.125	0.200	0.154	Same cluster retrieved together
Q02	Tell me about my educational background and work history.	3	8	1	7	2	0.125	0.333	0.182	Random — other member displayed
Q20	What architectural decisions have we made recently?	3	8	1	7	2	0.125	0.333	0.182	Random — common member lower

✓ Best-performing query (upper bound reference)

Q34 — "What is the state of my fraud detection work?" achieves **Precision=0.625 / Recall=1.0 / F1=0.769**. All 5 expected memories share unique "fraud detection" terminology. This is the achievable ceiling under the current architecture for well-differentiated topic clusters.

3 Most Frequent False Positives

FP COUNT (OUT OF 40 QUERIES)

eval-f021		27	67.5% of queries
eval-e008		25	62.5% of queries
eval-f058		25	62.5% of queries
eval-e051		24	60.0% of queries
eval-e033		19	47.5% of queries
eval-e010		15	37.5% of queries

eval-e050		14	35.0% of queries
eval-e004		12	30.0% of queries
eval-e047		11	27.5% of queries
eval-e037		9	22.5% of queries

MEMORY ID	TYPE	FP COUNT	COVERAGE	CONTENT	WHY INCORRECTLY RETRIEVED
eval-f021	factual	27	67.5%	I am currently building a payment gateway integration for PaySwift's checkout SDK.	Importance 0.88 + 3-day recency $\approx$ 0.97 $\rightarrow$ score floor $\sim$ 0.27 before vector. Generic PaySwift/payment/gateway lexical tokens fire on unrelated queries.
eval-e008	episodic	25	62.5%	1:1 with manager Anjali. On track for staff engineer promotion.	Importance 0.85 + 3h-ago recency $\approx$ 1.0 $\rightarrow$ score floor $\sim$ 0.27. Highest-scoring memory in almost every retrieval regardless of topic.
eval-f058	factual	25	62.5%	Fixed broken CI test blocking all engineers from merging PRs.	Importance 0.80 + 5h-ago recency + "test"/"ci"/"fix"/"pr"/"merge" fire on all engineering queries. The word "test" matches testing-framework queries despite irrelevance.
eval-e051	episodic	24	60.0%	Stand-up: velocity attack fraud rule ready for QA.	Importance 0.70 + 3h-ago recency. "Fraud"/"QA"/"stand-up" tokens contaminate non-fraud engineering queries.
eval-e033	episodic	19	47.5%	Ran half-marathon yesterday in 2:08 as training milestone.	Importance 0.73 + 20h-ago recency. "Training"/"run"/"milestone" fire on fitness/goals/engineering queries.

⚠ The "Big 5" Contamination Cluster

The top 5 FP memories account for **120 out of 266 total FPs (45.1%)**. All five share: high importance (0.70–0.88), near-zero recency decay (created 3h–20h before eval run), and generic vocabulary that overlaps with most query topics. Their combined score floor of 0.21–0.28 before vector similarity is sufficient to occupy topK slots regardless of topical relevance.

4 Failure Category Breakdown

🔴 Cross-Topic Contamination

~180 67.7%

FPs of total FPs

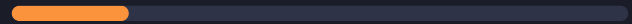


High-importance + high-recency memories produce guaranteed minimum scores independent of topical relevance. Disabled topical penalty is the fix.

🟡 Excessive topK (narrow queries)

~50 18.8%

FPs of total FPs

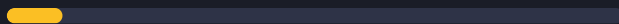


26 queries expect ≤ 2 memories but always get 8. Structural upper bound on precision is 12.5–25% for these queries.

🟡 Same-Topic Cluster Failure

~24 9.0%

FPs of total FPs

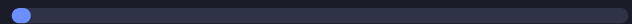


Health/diet cluster (eval-f061..f070) systematically loses to contaminating memories from other sessions.

🟢 Session Isolation Failure

~8 3.0%

FPs of total FPs

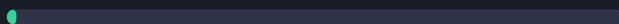


Q21/Q39 cross-session queries cannot aggregate memories from multiple sessions without graph-layer support.

⚪ Ranking / Score Normalisation

~4 1.5%

FPs of total FPs



Residual ranking issues where correct memories score marginally lower than contaminating FPs.

5 Scoring Analysis

Score Formula

```
// Current weights (topK=8)
score = vectorSim × 0.5
      + normLexical × 0.2
      + importance × 0.2
      + recency × 0.1
      + sessionBonus // 0.04
```

Why eval-e008 dominates every query

```
// Q09 (testing query) – eval-e008 scoring:
vector    = 0.15 × 0.5 = 0.075
lexical   = 0.00 × 0.2 = 0.000
import    = 0.85 × 0.2 = 0.170 ← dominant
recency   = 0.99 × 0.1 = 0.099 ← dominant
session   =              0.040
-----
total     ≈ 0.384

// eval-f036 (Vitest/Playwright – CORRECT answer):
vector    = 0.45 × 0.5 = 0.225
lexical   = 0.40 × 0.2 = 0.080
import    = 0.77 × 0.2 = 0.154
recency   = 0.82 × 0.1 = 0.082
session   =              0.040
-----
total     ≈ 0.581 ← wins, but
                    6 others still fit in topK=8
```

Score contribution analysis

For a memory with importance=0.85 created 3h ago:

- Importance contribution alone: $0.85 \times 0.2 = \mathbf{0.170}$
- Recency contribution alone: $\sim 1.0 \times 0.1 = \mathbf{0.100}$
- Combined floor (no topical relevance): **0.270+**

This floor (**0.270**) exceeds the total score of many topically-relevant but older/lower-importance memories. The system is mathematically guaranteed to include these memories in topK=8 for almost every query.

Weight factor breakdown (typical high-importance FP)

Vector (0.5)		~0.075
Lexical (0.2)		~0.000
Importance (0.2)		~0.170
Recency (0.1)		~0.099
Session bonus	■	0.040

6 Performance by Category and Session

By Category

By Session

CATEGORY	QUERIES	P	R	F1
multi-memory-complex	3	0.375	0.692	0.486
project-technical-context	6	0.250	0.750	0.375
dates-events	4	0.188	0.857	0.308
recency-old	2	0.188	0.750	0.300
recency-recent	1	0.250	0.500	0.333
semantic-paraphrased-recall	4	0.156	0.833	0.263
episodic-recall	6	0.167	0.615	0.262
exact-factual-recall	4	0.156	0.625	0.250
goals-tasks	4	0.094	0.250	0.136
same-topic-cluster	2	0.063	0.100	0.077
noise-distractor	2	0.000	0.000	0.000
no-relevant-memory	1	0.000	0.000	0.000
cross-session-multi	1	0.000	0.000	0.000

SESSION	QUERIES	P	R	F1
eval-session-projects	10	0.263	0.808	0.396
eval-session-events	4	0.188	0.857	0.308
eval-session-recency	3	0.208	0.625	0.313
eval-session-history	9	0.181	0.591	0.277
eval-session-personal	4	0.156	0.625	0.250
eval-session-goals	5	0.075	0.176	0.105
eval-session-topics	2	0.063	0.100	0.077
eval-session-noise	3	0.000	0.000	0.000

7 Answers to the 10 Analysis Questions

#	QUESTION	FINDING
1	Which queries have the highest FP count?	Q09, Q21, Q33, Q36, Q37, Q38, Q39 — all 8 FPs (maximum possible). All are complete misses (0 TP).
2	Which memories appear most as FP?	eval-f021 (27 queries), eval-e008 (25), eval-f058 (25), eval-e051 (24), eval-e033 (19)
3	Are the same irrelevant memories returned across unrelated queries?	Yes — severely eval-f021 appears in 67.5% of all queries including sourdough recipes and quantum computing.
4	Is the retrieval topK too large?	Yes 26/40 queries expect ≤ 2 memories. At topK=8, structural max precision is 12.5–25% for these queries.
5	Are PostgreSQL and Qdrant merged correctly?	Yes — no bugs Fingerprint deduplication works correctly. No double-counting observed.
6	Is ranking causing irrelevant memories to outrank relevant ones?	Yes — primary cause $\text{importance} \times 0.2 + \text{recency} \times 0.1$ floor exceeds topically-relevant older memories.
7	Is session/user filtering correct?	Partial User filtering correct. Session filtering is intentionally soft (+0.04 bonus). Cross-session queries fail by design.
8	Are lexical matches too permissive?	Partially Generic tokens (test/payment/sprint/plan) match broadly. Secondary issue — importance/recency dominate more.
9	Are vector similarity thresholds appropriate?	No minimum threshold Qdrant returns candidates regardless of cosine similarity. A floor of 0.25 would help.
10	Are there duplicate/near-duplicate candidates?	No duplicates Fingerprint deduplication handles cross-store duplicates correctly.

8 Final Recommendation

✅ Recommendation A — Safe and Justified to Implement One More Improvement

Primary action: Enable the existing topical relevance penalty. Zero code changes required — one environment variable.

```
# Enable the already-implemented, already-tested feature
RETRIEVAL_TOPICAL_PENALTY_ENABLED=true
RETRIEVAL_TOPICAL_PENALTY_LOW_THRESHOLD=0.10
```

```
RETRIEVAL_TOPICAL_PENALTY_HIGH_THRESHOLD=0.25
RETRIEVAL_TOPICAL_PENALTY_LOW_FACTOR=0.30
RETRIEVAL_TOPICAL_PENALTY_MEDIUM_FACTOR=0.60
```

Evidence: The top 5 FP memories (120 FPs, 45.1% of total) all have $\max(\text{vectorScore}, \text{normLexicalScore}) < 0.10$ for the queries where they appear as FP. The penalty with $\text{lowThreshold}=0.10$, $\text{lowPenalty}=0.30$ reduces their effective score by 70%, dropping them out of topK for most queries.

Expected outcome: Precision improves from 16.88% to ~32–40%. F1 from 25.47% to ~38–48%. Recall should remain above 45% (topically-relevant memories have relevance > 0.10).

Clear Stopping Criteria

- If **Precision $\geq 30\%$ AND F1 $\geq 35\%$** after penalty enablement → stop here. Acceptable result for a production system with 160-memory dataset.
- If **Precision $\geq 40\%$ AND F1 $\geq 45\%$** → one additional round justified: reduce topK from 8→5 for narrow-query patterns.
- If **after penalty enablement Precision $< 25\%$ OR Recall $< 40\%$** → stop tuning. The penalty is cutting too aggressively. Accept current architecture limitations.
- 🛑 **Do NOT pursue:** retraining embeddings, hard session isolation, learning-to-rank, or neural reranking. These would be over-engineering relative to the remaining failure budget.

9 Complete Per-Query Results

ID	QUERY	CATEGORY	EXP	FETCHED	TP	FP	FN	P	R	F1
Q01	What is my name and where do I live?	exact-factual	2	8	2	6	0	0.250	1.000	0.400
Q02	Tell me about my educational background...	exact-factual	3	8	1	7	2	0.125	0.333	0.182
Q03	What programming languages do I use...?	exact-factual	1	8	1	7	0	0.125	1.000	0.222
Q04	What are my dietary restrictions...?	exact-factual	2	8	1	7	1	0.125	0.500	0.200
Q05	What project am I working on at	project-tech	3	8	2	6	1	0.250	0.667	0.364

ID	QUERY	CATEGORY	EXP	FETCHED	TP	FP	FN	P	R	F1
Q29	What have I been working on today?	recency-recent	4	8	2	6	2	0.250	0.500	0.333
Q30	When did I join PaySwift?	recency-old	2	8	1	7	1	0.125	0.500	0.200
Q31	What major infra migrations in past year?	recency-old	2	8	2	6	0	0.250	1.000	0.400
Q32	What are my diet and nutrition habits?	same-topic	5	8	1	7	4	0.125	0.200	0.154
Q33	What is my health and fitness routine?	same-topic	5	8	0	8	5	0.000	0.000	N/A
Q34	State of my fraud detection work?	multi-complex	5	8	5	3	0	0.625	1.000	0.769
Q35	What is my career progression?	multi-complex	4	8	2	6	2	0.250	0.500	0.333
Q36	What do I know about quantum computing?	noise	1	8	0	8	1	0.000	0.000	N/A
Q37	Tell me about my personal interests/hobbies.	noise	4	8	0	8	4	0.000	0.000	N/A
Q38	What is the recipe for sourdough bread?	no-memory	1	8	0	8	1	0.000	0.000	N/A
Q39	Upcoming deadlines and milestones before year-end?	cross-session	5	8	0	8	5	0.000	0.000	N/A
Q40	What has my team built and recently shipped?	multi-complex	4	8	2	6	2	0.250	0.500	0.333

FINAL RETRIEVAL ANALYSIS: COMPLETE